6

Pemetaan Diagram (E)ER ke Skema Relasional

CSGE602070 - BASIS DATA

Acknowledgements

This slide is a modification to supplementary slide of
“Database System”, 7th edition, Elmasri/Navathe, 2015: **Chapter 9** used in “Basis Data” course in academic years 2018/2019 in the Faculty of Computer Science, Universitas Indonesia.

Additional resources: presentation prepared by Prof Steven A. Demurjian, Sr
(http://www.engr.uconn.edu/~steve/courses.html

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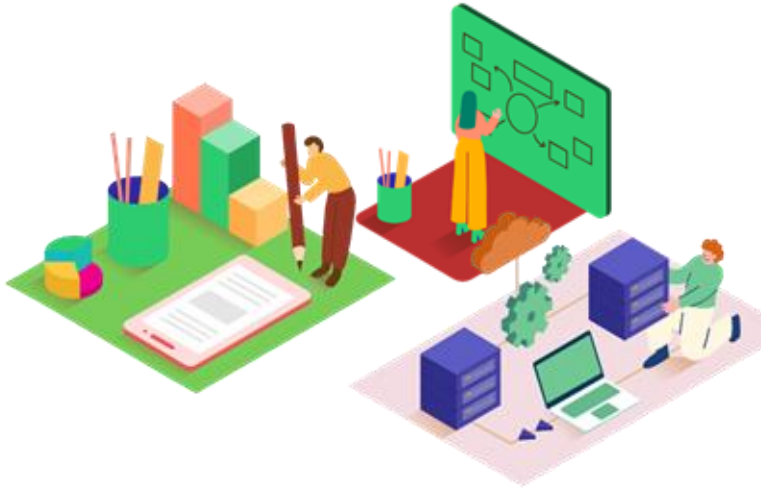
Tujuan Pembelajaran

Setelah mengikuti pembelajaran pada topik ini, Anda diharapkan dapat **memetakan diagram ER atau EER ke Skema Relasional**

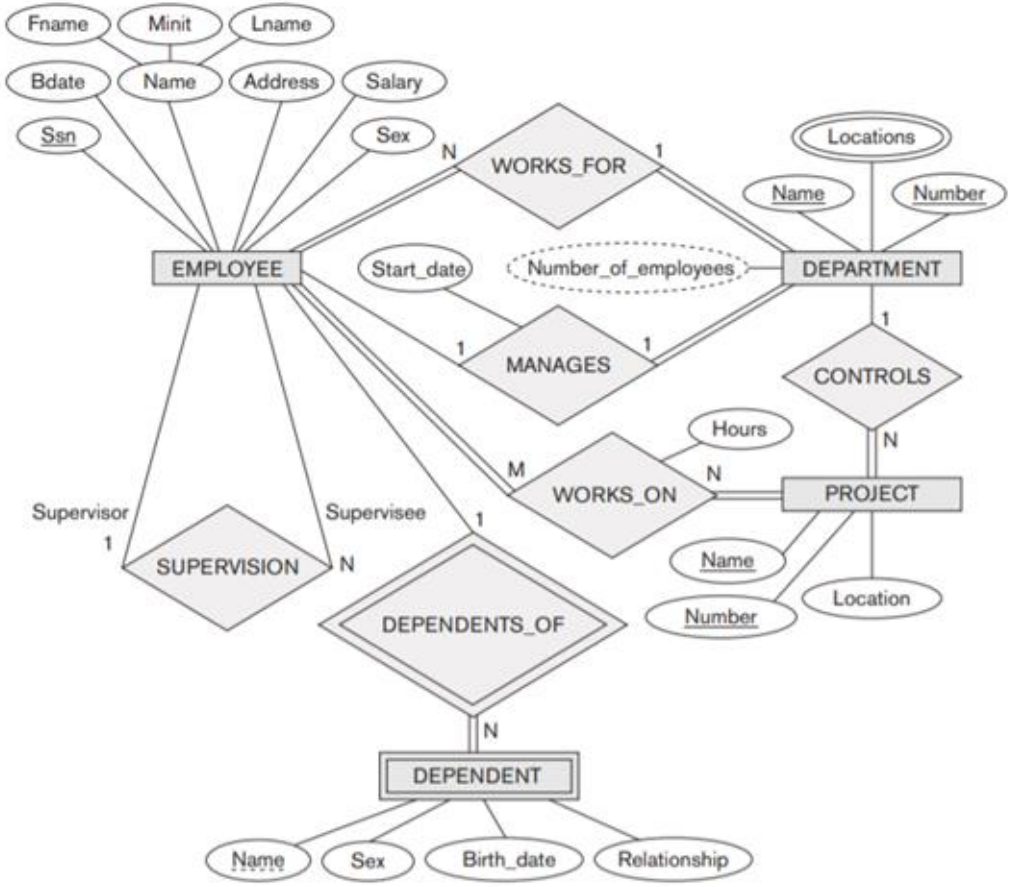
Outline

1. Pemetaan Model ER ke Skema Relasional

2. Pemetaan Model EER ke Skema Relasional



ER Diagram


Fname
Minit
Lname
Bdate
Name
Address
Salary
Ssn
Sex
N
WORKS_FOR
1
Locations
Name
Number
EMPLOYEE
Start_date
Number_of_employees
DEPARTMENT
1
MANAGES
1
1
CONTROLS
Hours
N
M
WORKS_ON
N
PROJECT
Supervisor
Supervisee
1
1
SUPERVISION
N
Name
Location
Number
DEPENDENTS_OF
N
DEPENDENT
Name
Sex
Birth_date
Relationship

Skema Relasional

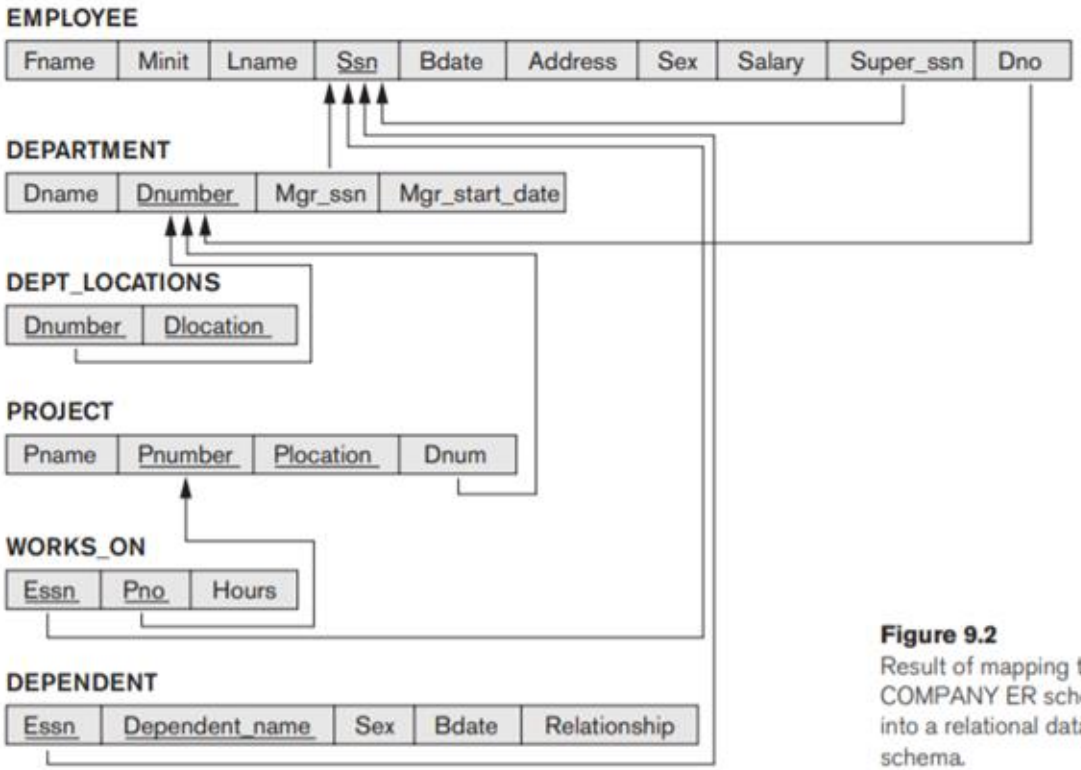


Figure 9.2
Result of mapping the COMPANY ER schema into a relational database schema.

Algoritma Pemetaan Diagram ER ke Model Relasional

Urutan **tahapan** pemetaan diagram ER ke Model Relasional

1. Pemetaan **Regular Entity Types**
2. Pemetaan **Weak Entity Types**
3. Pemetaan **Binary 1:1 Relationship Types**
4. Pemetaan **Binary 1:N Relationship Types**
5. Pemetaan **Binary M:N Relationship Types**
6. Pemetaan **Multivalued attributes**
7. Pemetaan **N-ary (Higher Degree) Relationship Types**

1. Pemetaan *Regular Entity Types*

EMPLOYEE

SSN	FName	MInit	LName	BDate	Address	Sex	Salary

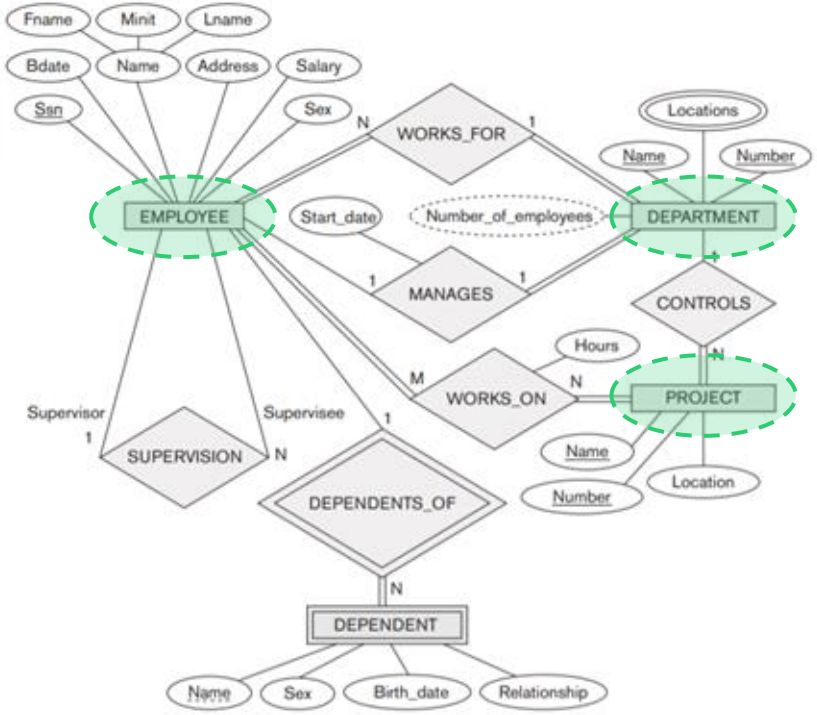
DEPARTMENT

Dname	Dnumber

PROJECT

Pname	Pnumber	Plocation

Map each **regular entity type** to a **relation**



2. Pemetaan *Weak Entity Types*

EMPLOYEE

SSN	FName	MInit	LName	BDate	Address	Sex	Salary

DEPARTMENT

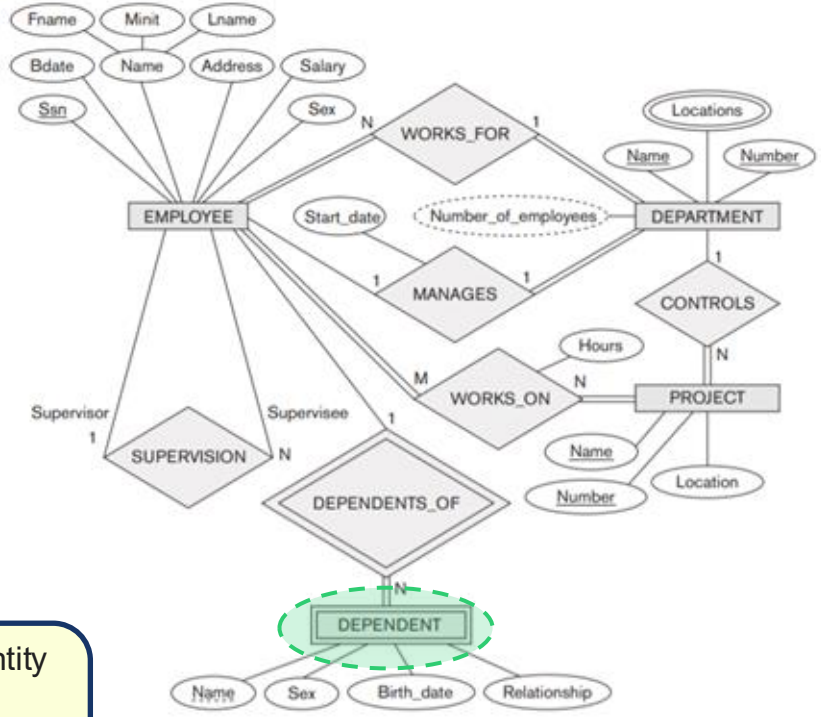
Dname	Dnumber

PROJECT

Pname	Pnumber	Plocation

DEPENDENT

ESsn	Dependent_Name	BirthDate	Sex	Relationship



Map weak entity type to a relation, copy the key of its owner entity type, and combine it with its partial key as the relation key.

Draw an arrow from the copied key to key of the owner entity to depict the referential integrity constraint.

3. Pemetaan Binary 1-1 Relationship Types (1)

EMPLOYEE

SSN	FName	MInit	LName	BDate	Address	Sex	Salary

DEPARTMENT

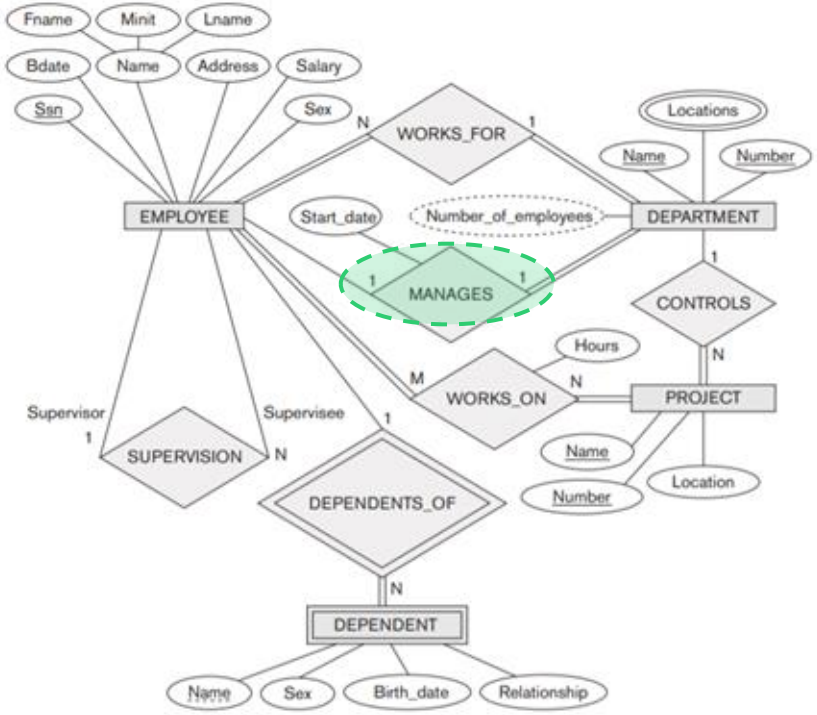
Dname	Dnumber	Mgr_Ssn	Mgr_Start_Date

PROJECT

Pname	Pnumber	Plocation

DEPENDENT

ESsn	Dependent_Name	BirthDate	Sex	Relationship



3. Pemetaan Binary 1-1 Relationship Types (2)

Tiga **pendekatan** dalam pemetaan:

1. Pendekatan **foreign key**
   - ➔ Pilih salah satu relasi (misal S), masukkan primary key dari relasi T sebagai foreign key pada S.
   - ➔ Relasi S yang dipilih sebaiknya yang berpartisipasi **total**.

2. Opsi **penggabungan relasi**
   - ➔ Kedua entity types dan relationship digabungkan menjadi satu relasi.
   - ➔ Sesuai jika partisipasi **kedua entity types dalam relationship tersebut bersifat total**.

3. Opsi **cross-reference / relationship relation**
   - ➔ Membuat **relasi baru** yang mengandung primary key dari kedua entity types yang terhubung dalam relationship tersebut.
   - ➔ Dapat dilakukan pada **berbagai kondisi** (partisipasi total maupun parsial)

4. Pemetaan Binary 1-N Relationship Types

EMPLOYEE

SSN	FName	MInit	LName	BDate	Address	Sex	Salary	Super_Ssn	Dno

DEPARTMENT

Dname	Dnumber	Mgr_Ssn	Mgr_Start_Date

PROJECT

Pname	Pnumber	Plocation	Plocation	DNum

DEPENDENT

ESsn	Dependent_Name	BirthDate	Sex	Relationship

Copy the key of the entity that connects to the “1” side of the 1:N to the entity on the “N” side of the 1:N.

Draw an arrow from copied key in the “N” side entity to the key in the “1” side entity to depict the referential integrity constraint.

5. Pemetaan Binary M-N Relationship Types

EMPLOYEE

SSN	FName	MInit	LName	BDate	Address	Sex	Salary	Super_Ssn	Dno

DEPARTMENT

Dname	Dnumber	Mgr_Ssn	Mgr_Start_Date

PROJECT

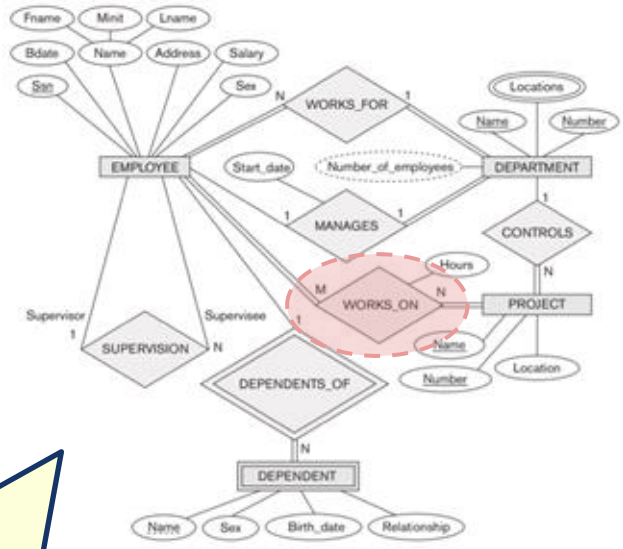
Pname	Pnumber	Plocation	Plocation	DNum

DEPENDENT

ESsn	Dependent_Name	BirthDate	Sex	Relationship

WORKS_ON

ESsn	PNo	Hours



Create a new relation, copy the keys from entities of the M:N relationship, combined them as the key of the new relation, and add the relationship attribute, *if any*.

Draw two arrows, each from the copied keys to the original relation to depict the referential integrity constraint.

6. Pemetaan Multivalued Attribute

DEPARTMENT

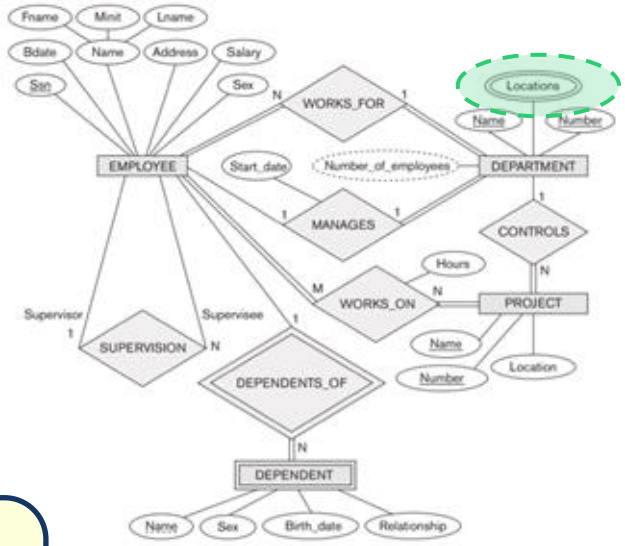
Dname	Dnumber	Mgr_Ssn	Mgr_Start_Date

DEPT_LOCATIONS

Dnumber	DLocation

Create a new relation for the multivalued attribute, add the key of the relation having that multivalued attribute, combine the added key with the mulivalued attribute as the key of the new relation.

Draw an arrow from the copied key in the new relation to the original relation to depict the referential integrity constraint.



Hasil Pemetaan Diagram ER COMPANY

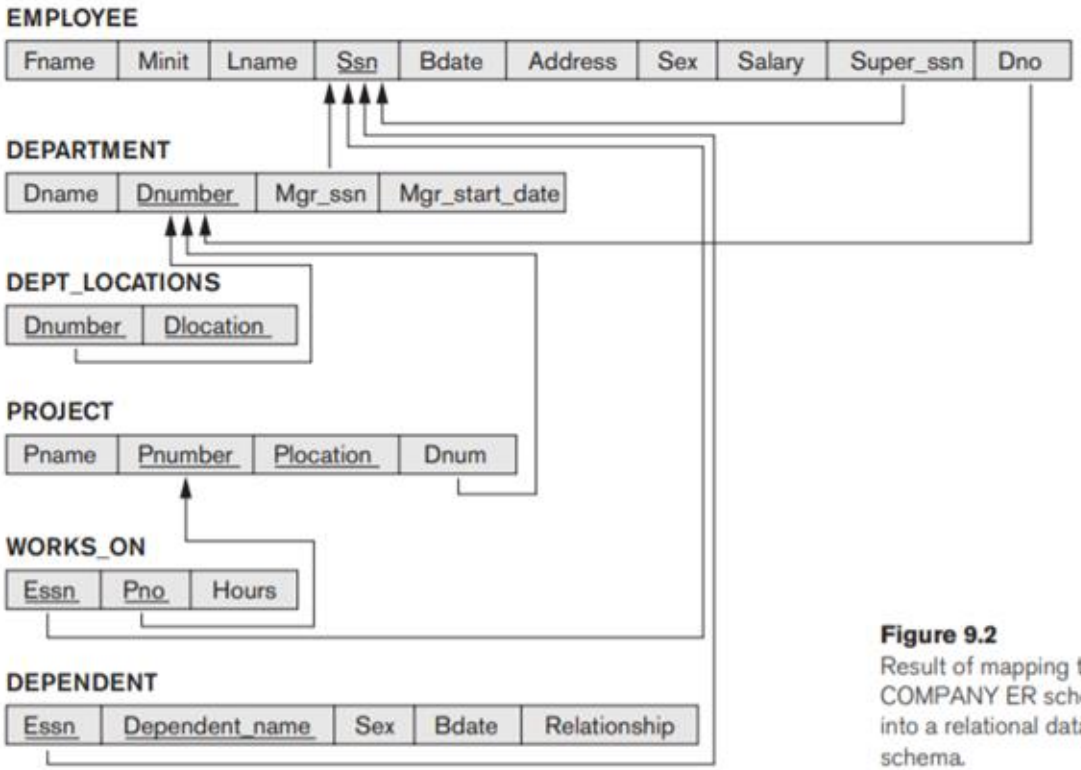
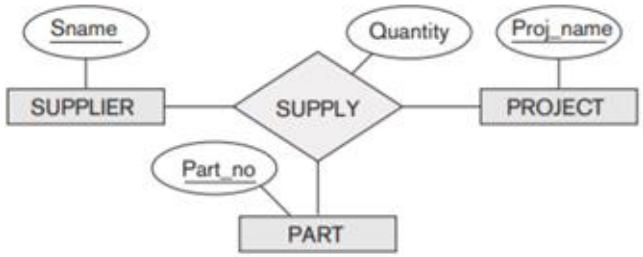
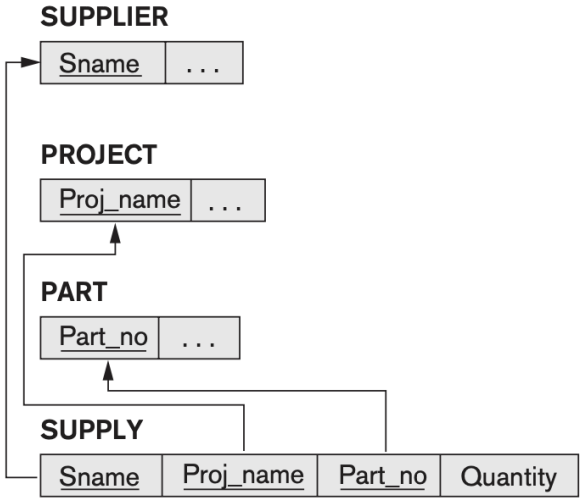


Figure 9.2
Result of mapping the COMPANY ER schema into a relational database schema.

7. Pemetaan Higher Degree Relationship Types





Map the higher degree relationship type into a new relation, and then copy each key of the participating entities into the new relation.

Draw an arrow from each copied key in the new relation to the original relation to depict the referential integrity constraint.

Add the relationship attribute to the new relation, *if any*.

Pemetaan Derived Attribute

Opsi 1**: tidak disimpan** pada basis data

- ➔ Nilai dari derived attribute dihitung pada saat diperlukan (perlu waktu komputasi).
- ➔ Menghemat storage.
- ➔ Konsistensi antara nilai derived attribute dengan stored attribute yang berkaitan dapat dijaga.

Opsi 2: **disimpan** pada basis data

- ➔ Derived attribute diperlakukan sebagaimana simple attribute.
- ➔ Untuk menjamin konsistensi, nilai dari derived attribute ini sebaiknya jangan dimasukkan oleh operator, namun dilakukan secara otomatis oleh sistem yang kita buat.

Latihan (1)

Petakan diagram di bawah ini ke skema relasional

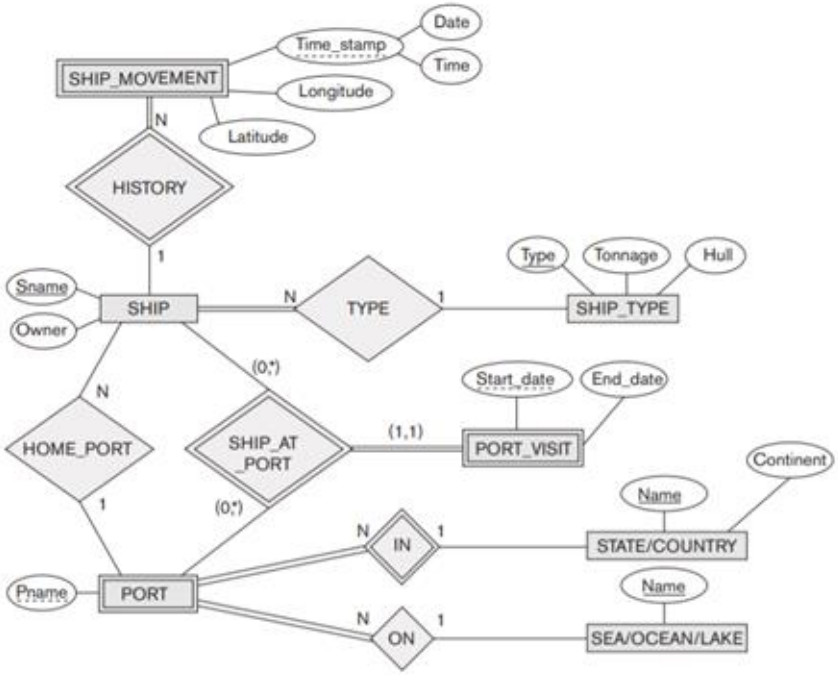


Figure 9.8
An ER schema for a SHIP_TRACKING database.

Latihan (2)

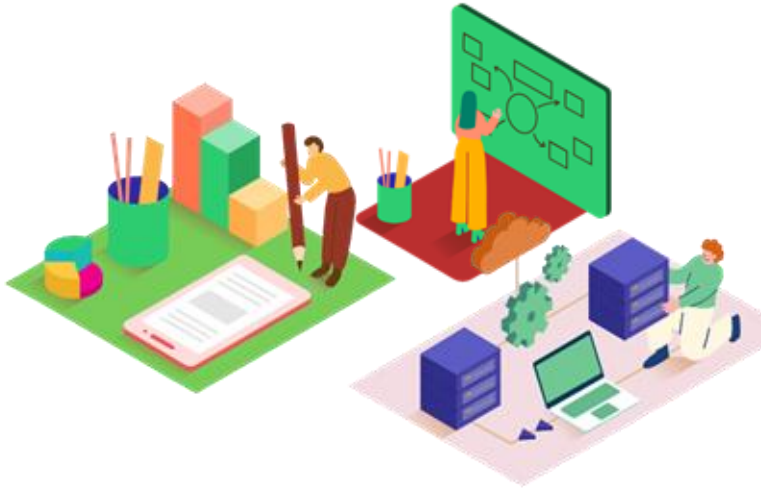
1. Menurut Anda, apa dampaknya jika suatu binary **M-N relationship type tidak dipetakan** menjadi relasi baru, tapi dipetakan seperti 1-N relationship type?

1. Apakah kita dapat mengubah urutan langkah 1 – langkah 7 pemetaan ER ke skema relasional, namun memberikan hasil pemetaan yang sama?

Outline

1. Pemetaan Model ER ke Skema Relasional

2. Pemetaan Model EER ke Skema Relasional



Pemetaan Model EER ke Skema Relasional

Pemetaan spesialisasi dan generalisasi

Pemetaan shared subclasses

Pemetaan union types

Pemetaan Spesialiasi dan Generalisasi

Pilihan pemetaan spesialisasi dan generalisasi:

Opsi 1. Banyak relasi: superclass dan subclasses

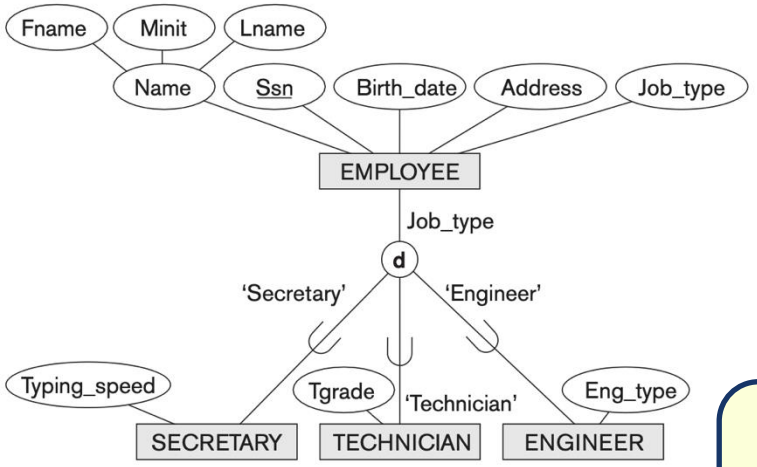
Opsi 2. Banyak relasi: subclasses saja

Opsi 3. Satu relasi dengan satu type attribute

Opsi 4. Satu relasi dengan banyak type attribute

Opsi 1. Banyak relasi: Superclass dan Subclasses

Semua constraint: OK



Map each subclass into a new relation

EMPLOYEE

Name	MInit	LName	Ssn	Birth_date	Address	Job_type

SECRETARY

Ssn	Typing_speed

TECHNICIAN

Ssn	Technician

ENGINEER

Ssn	Eng_type

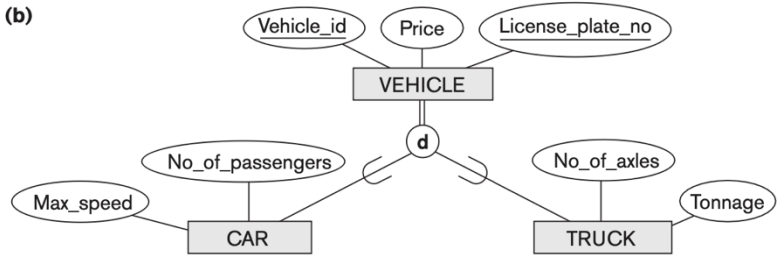
Copy the superclass' key attribute to its subclass relations

Draw an arrow from the subclass' key to its superclass's key to depict referential integrity constraint

Opsi 2. Banyak Relasi: Hanya Subclasses

Hanya untuk: **Total** disjoin atau **Total** overlap

Superclass' attributes are mapped to the subclass relation's attributes



CAR

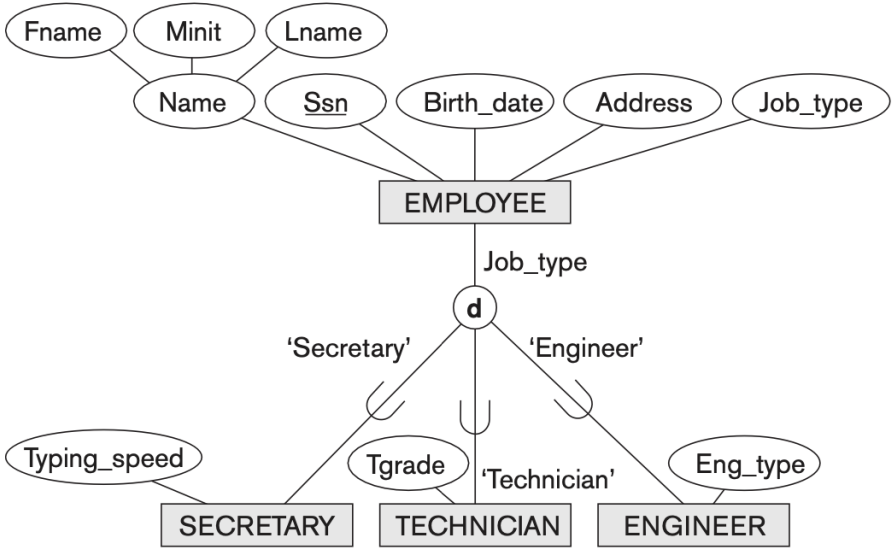
Vehicle_id	License_plate_no	Price	Max_speed	No_of_passengers

TRUCK

Vehicle_id	License_plate_no	Price	No_of_axles	Tonnage

Opsi 3. Satu Relasi dengan Satu *Type Attribute*

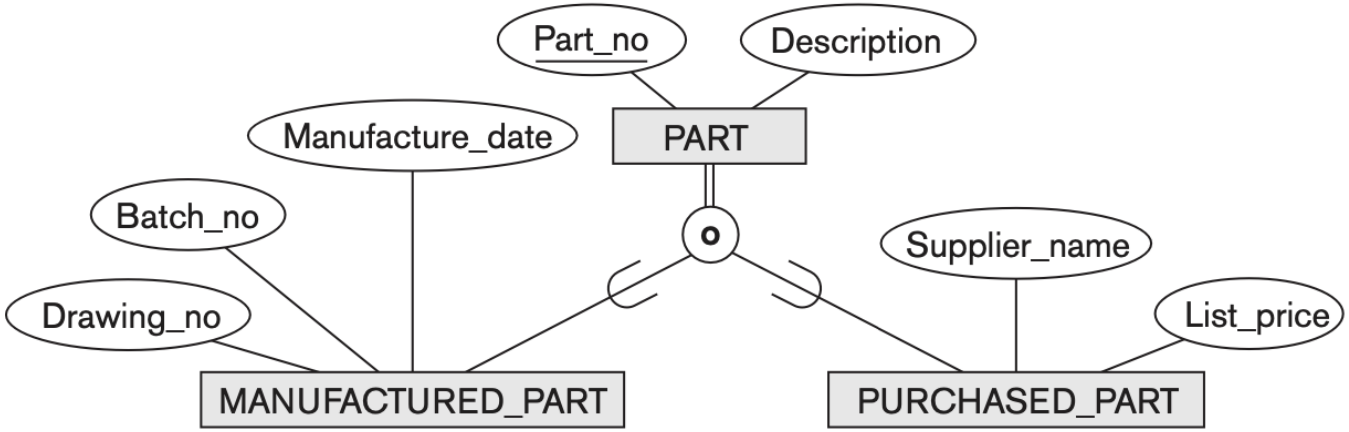
Hanya untuk: Total **disjoin** atau Parsial **disjoin**



EMPLOYEE

Name	MInit	LName	Ssn	Birth_date	Address	Job_type	Typing_speed	Tgrade	Eng_type

Opsi 4. Satu Relasi dengan Banyak *Type Attribute*



PART

Part_no	Description	Mflag	Drawing_no	Batch_no	Manufacture_date	Pflag	Supplier_name	List_price

Pemetaan Shared Subclasses

Pemetaan shared subclasses dapat memilih opsi 1 sampai opsi 4 yang dijelaskan sebelumnya.

PERSON

Ssn	Name	Sex	Address	Birth_date

EMPLOYEE

Ssn	Salary	Employee_ type	Position	Rank	Percent Time	RA Flag	TA Flag	Project	Course

ALUMNUS

Ssn

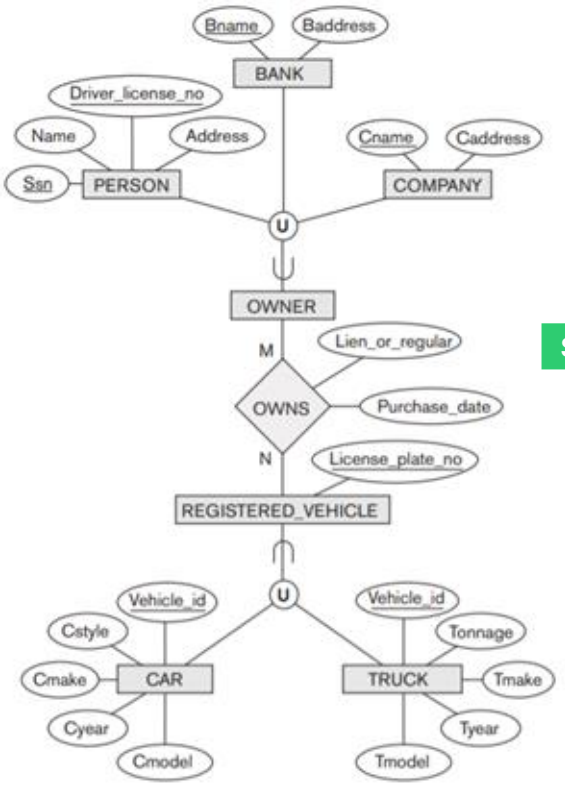
ALUMNUS_DEGREES

Ssn	Year	Degree	Major

STUDENT

Ssn	Major Dept	GradFlag	Undergrad Flag	DegreeProgram	Class	StudAssistFlag

Pemetaan Union Types



Surrogate Key

PERSON

Ssn	Driver_license_no	Name	Address	Owner_id

BANK

Bname	Baddress	Owner_id

COMPANY

Cname	Caddress	Owner_id

OWNER

Owner_id

REGISTERED_VEHICLE

Vehicle_id	License_plate_number

CAR

Vehicle_id	Cstyle	Cmake	Cmodel	Cyear

TRUCK

Vehicle_id	Tmake	Tmodel	Tonnage	Tyear

OWNS

Owner_id	Vehicle_id	Purchase_date	Lien_or_regular

The superclasses have the same key (Vehicle_id) → **no need for surrogate key**

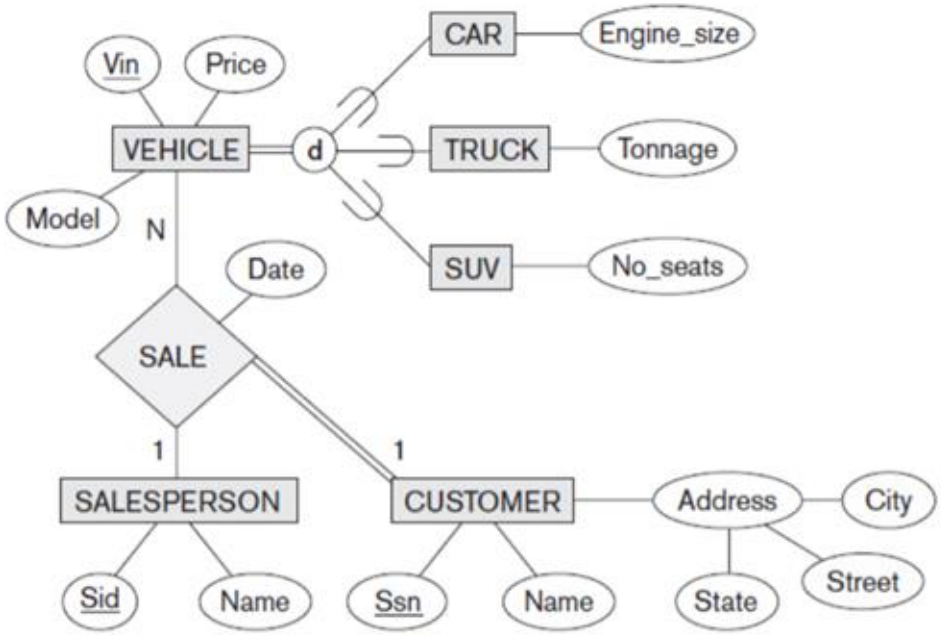
Rangkuman

Tahapan pemetaan model (E)ER ke skema relasional:

1. **Pemetaan regular entity types**
2. **Pemetaan weak entity types**
3. **Pemetaan binary 1:1 relation types**
4. **Pemetaan binary 1:N relationship types**
5. **Pemetaan binary M:N relationship types**
6. **Pemetaan multivalued attributes**
7. **Pemetaan N-ary relationship types**
8. **Pemetaan spesialisasi dan generalisasi**
9. **Pemetaan shared subclasses**
10. **Pemetaan union types**

Latihan

Petakan diagram di bawah ini ke skema relasional



Q&A